Fairness and Bias in Machine Learning

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Logistic Regression

- The dependent variable (Y) is a Bernoulli random variable
  - Y = +1 – "passing the bar exam"
  - Y = -1 – "failing the bar exam"

- We seek to predict the probability of a success outcome of the dependent variable Y as a function of independent variables $x_1, x_2 .. x_k$

- We predict the *likelihood* that Y = +1 as follows:
  - $\Pr(Y = +1) = \frac{e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_k x_k}}{1+e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_k x_k}}$

This is guaranteed to be between 0 and 1

Logistic regression

Logistic regression model

$$\min_{\beta_0,\boldsymbol{\beta}} \sum_{i=1}^{n} \log\left(1+e^{-y_i(\boldsymbol{\beta}^\top x_i+\beta_0)}\right).$$

Parameters of a Logistic regression model

MIO model

$$\min_{\mathbf{z}\in\mathcal{Z}_{\tau_w,\tau_b}} \min_{\beta_0,\boldsymbol{\beta}} \sum_{i=1}^{n} \log\left(1+e^{-\underbrace{y_i(1-2z_i)}(\boldsymbol{\beta}^\top x_i+\beta_0)}\right).$$

Updated outcome label

Binary variables

MIO model (linearizing product terms)

$$\min_{\mathbf{z}} \min_{\boldsymbol{\beta},\boldsymbol{\gamma}} \quad f(\boldsymbol{\beta},\boldsymbol{\gamma}) := \sum_{i=1}^{n} \log\left(1 + e^{-y_i(\boldsymbol{\beta}^\top \mathbf{x}_i + \beta_0) + 2y_i(\boldsymbol{\gamma}_i^\top \mathbf{x}_i + \gamma_{i,0})}\right)$$

$$\text{s.t} \quad \sum_{i \in \mathcal{W}} z_i = \tau_w \cdot n_w,$$

label flips

$$\sum_{i \in \mathcal{B}} z_i = \tau_b \cdot n_b,$$

$$-z_i M_j \leq \gamma_{i,j} \leq z_i M_j, \; i \in [n], j \in [p],$$

Big-M constraints

$$-(1 - z_i) M_j \leq \gamma_{i,j} - \beta_j \leq (1 - z_i) M_j, \; i \in [n], j \in [p],$$

Implied constraints
(using binary variables)

$$\sum_{i \in \mathcal{W}} \gamma_{i,j} = \tau_w \cdot n_w \cdot \beta_j, \; j \in [p],$$

$$\sum_{i \in \mathcal{B}} \gamma_{i,j} = \tau_b \cdot n_w \cdot \beta_j, \; j \in [p],$$

$$z_i \in \{0, 1\}, i \in [n].$$

Law school students dataset

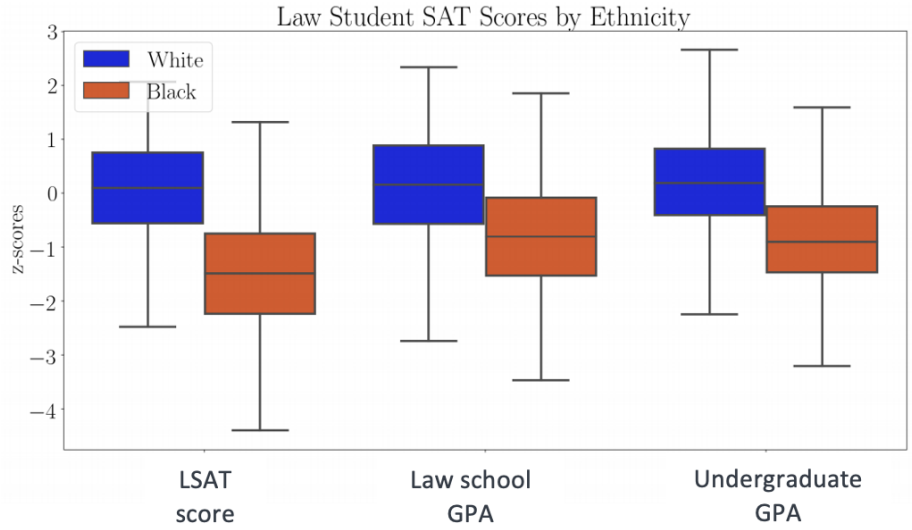
- This dataset originates from a study by Wightman (1998) that examined whether the bar exam (taken by law students in the US) is biased against ethnic minorities.

- It contains anonymized historical information including
  - Age,
  - LSAT scores (lsat),
  - First year law school GPA (zfygpa),
  - Cumulative law school GPA (zgpa), and
  - Undergraduate GPA (ugpa).

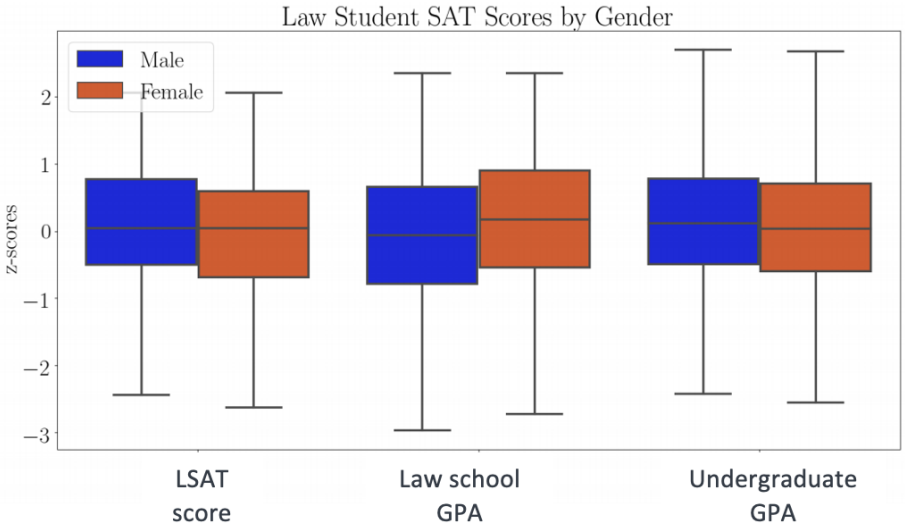
Law school students dataset

Figure 1 Distributions of LSAT scores, law school and undergraduate GPA by race and gender in the LSAC dataset ($\alpha_{\text{race}} = 0.3034$, $\alpha_{\text{gender}} = 0.0211$).

(a) Disparity across race labels.



(b) Parity across gender labels.



Law school students dataset

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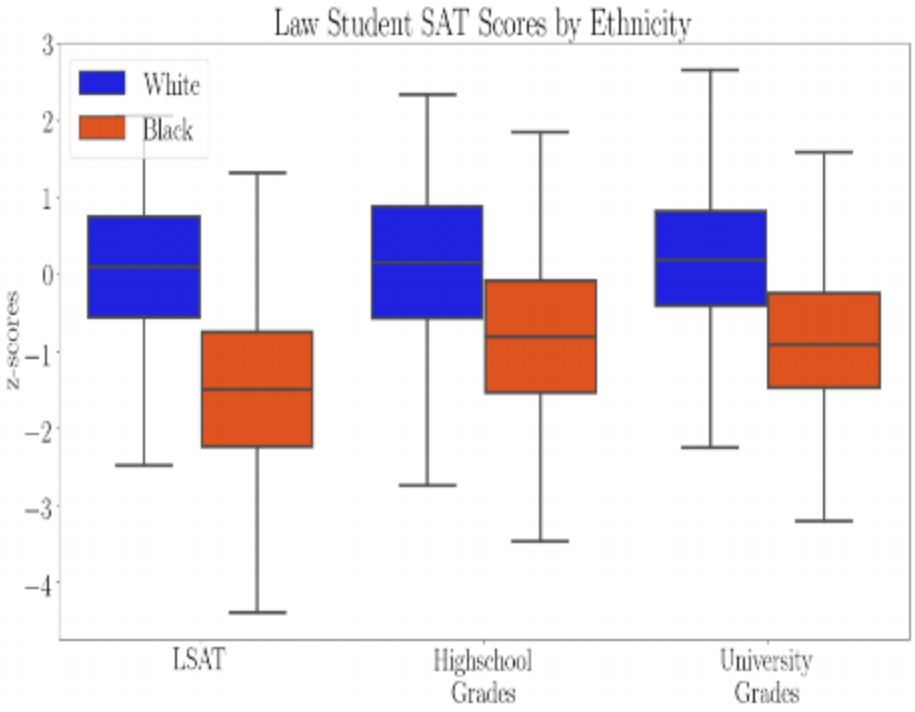
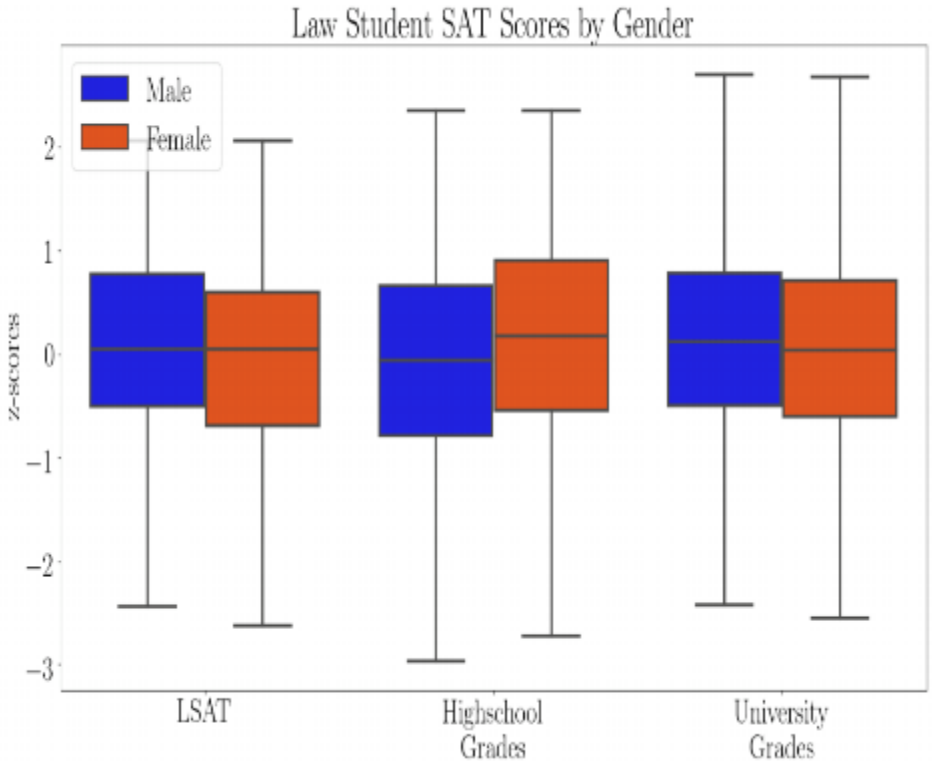




The Price of Diversity

Table 1 Trade-off between race parity and attributes of interest (reported in terms of z-scores of attributes: law school GPA, undergraduate GPA and LSAT score) in the LSAC dataset. DP-SVM, DP-LR correspond to Algorithm 1 applied to SVM and LR models respectively that achieve ϵ-demographic parity for race ($\alpha_{\text{race}} = 0.3034$, $\epsilon = 0.01$).

	White subpopulation					Black subpopulation				
Attribute	Data	SVM	DP-SVM	LR	DP-LR	Data	SVM	DP-SVM	LR	DP-LR
law school GPA	0.03	0.03	0.04	0.04	0.04	-0.76	-0.84	-0.92	-0.82	-0.88
UG GPA	0.05	0.05	0.06	0.06	0.06	-0.87	-0.81	-0.93	-0.85	-0.91
LSAT score	0.07	0.07	0.07	0.06	0.07	-1.46	-1.36	-1.47	-1.33	-1.47

Observations

- z-scores of attributes (GPA, LSAT scores) for admitted students among the white subpopulation *does not change significantly* even after employing MIO approach.

- However, the z-scores of attributes for admitted students among the black subpopulation *decreases slightly* signifying a *lowering of threshold* for passing the bar exam for black students that *helps to achieve demographic parity*.

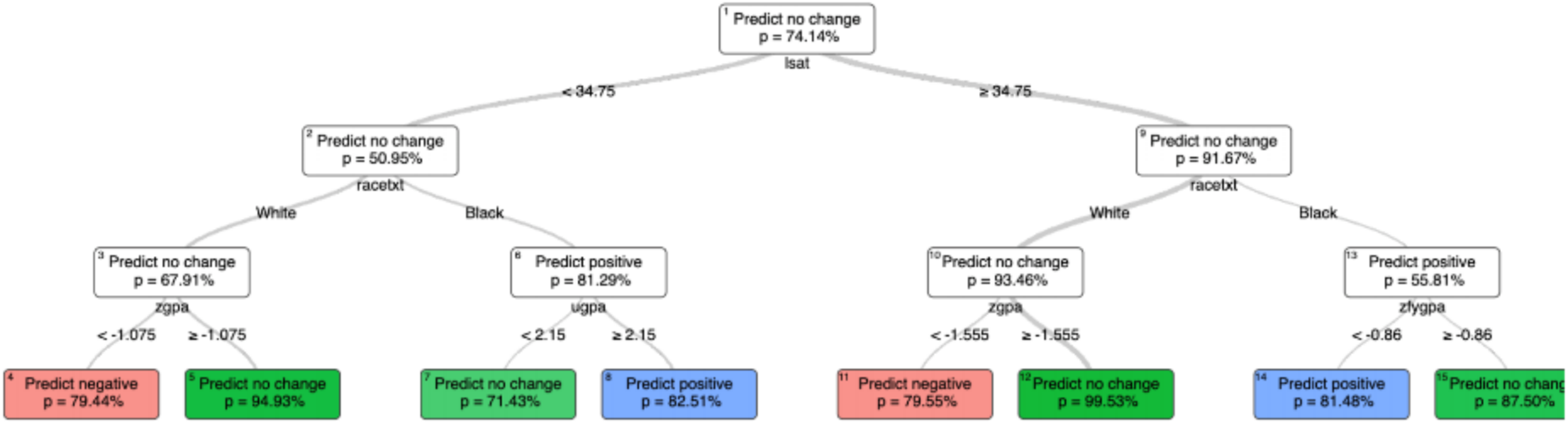
Actionable insights

- Train Optimal Classification Trees (OCTs) to understand *characteristics of observations* for which outcome labels are flipped after employing our MIO approach.

- Construct a dataset based on the results from MIO approach. Each observation is labeled as one of the following:
  - *positive* (outcome label changed to a positive label),
  - *negative* (outcome label changed to a negative label), or
  - *no change* (outcome label unchanged)

Actionable insights

Decision tree illustrating rules to achieve demographic parity for race in the law school students dataset. Attributes at each of the nodes: racetxt (race text), lsat (LSAT score), zgpa (Law school GPA) and ugpa (Undergraduate GPA). AUC: 0.67.



Observations

- The logistic regression model identifies criteria (based on GPA and LSAT scores) to select *most meritorious* students among the black subpopulation
  - (LSAT ≥ 34.75 & zfygpa < -0.86) or (LSAT < 34.75 & ugpa ≥ 2.15) who were *assigned a negative label* and flip their outcome to a *positive label*.

- The model also identifies criteria for the *least meritorious students* among the white subpopulation
  - (LSAT ≥ 34.75 & zgpa < -1.56) or (LSAT < 34.75 & zgpa < -1.08) who were *assigned a positive label* and flipped their outcome to a *negative label*.

Observations

- Leaf nodes that predict label *no change* for the outcome
  - Black subpopulation:
    - (LSAT ≥ 34.75 & zfygpa ≥ -0.86) – 93.4% of students *pass* the bar exam (*most qualified*).
    - (LSAT < 34.75 & ugpa < 2.15) – 76.2% of students *fail* the bar exam (*not qualified enough*).

  - White subpopulation:
    - (LSAT ≥ 34.75 & zgpa ≥ -1.56) – 78.7% of students *pass* the bar exam (*most qualified*)
    - (LSAT < 34.75 & zgpa < -1.08) – 87.3% of students *fail* the bar exam (*not qualified enough*).